PROBABILITY THEORY

Terminology

- A (stochastic) *experiment* is a procedure that yields one of a given set of possible outcomes
- The *sample space S* of the experiment is the set of possible outcomes.
- An event is a subset of sample space.
- A random variable is a function that assigns a real value to each outcome of an experiment

Random Variables

- A *"random variable"* V is any variable whose value is unknown, or whose value depends on the precise situation.
  - *E.g.*, the number of students in class today
  - Whether it will rain tonight (Boolean variable)
- The proposition $V=v_i$ may have an uncertain truth value, and may be assigned a *probability*.

Example

- A fair coin is flipped 3 times. Let S be the sample space of 8 possible outcomes, and let X be a random variable that assignees to an outcome the number of heads in this outcome.

- **Random variable X is a function** $X{:}S \rightarrow X(S)$,
  where $X(S)=\{0, 1, 2, 3\}$ is the range of X, which is the number of heads, and
  $S=\{$ (TTT), (TTH), (THH), (HTT), (HHT), (HHH), (THT), (HTH) $\}$

- X(TTT) = 0
  X(TTH) = X(HTT) = X(THT) = 1
  X(HHT) = X(THH) = X(HTH) = 2
  X(HHH) = 3

- The **probability distribution of random variable** X is given by
  P(X=3) = 1/8, P(X=2) = 3/8, P(X=1) = 3/8, P(X=0) = 1/8.

Experiments & Sample Spaces

- A (stochastic) *experiment* is any process by which a given random variable V gets assigned some *particular* value, and where this value is not necessarily known in advance.
  - We call it the "actual" value of the variable, as determined by that particular experiment.
- The *sample space* S of the experiment is just the domain of the random variable, $S = \mathrm{dom}[V]$.
- The *outcome* of the experiment is the specific value v_i of the random variable that is selected.

Events

- An *event* E is any set of possible outcomes in S…
  - That is, $E \subseteq S = \mathbf{dom}[V]$.
    - E.g., the event that "less than 50 people show up for our next class" is represented as the set $\{1, 2, \ldots, 49\}$ of values of the variable V = (# of people here next class).
- We say that event E *occurs* when the actual value of V is in E, which may be written $V \in E$.
  - Note that $V \in E$ denotes the proposition (of uncertain truth) asserting that the actual outcome (value of V) will be one of the outcomes in the set E.

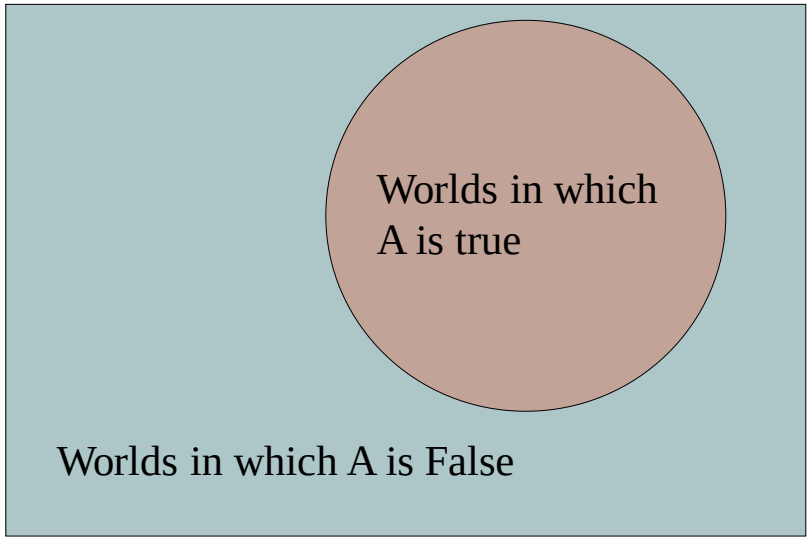
Probabilities

We write P(A) as “the fraction of possible worlds in which A is true”

Visualizing A

Event space of all possible worlds

Its area is 1



P(A) = Area of reddish oval

Probability

- The *probability* $p = \Pr[E] \in [0,1]$ of an event E is a real number representing our degree of certainty that E will occur.
  - If $\Pr[E] = 1$, then E is absolutely certain to occur,
    - thus $V \in E$ has the truth value **True**.
  - If $\Pr[E] = 0$, then E is absolutely certain *not* to occur,
    - thus $V \in E$ has the truth value **False**.
  - If $\Pr[E] = ½$, then we are *maximally uncertain* about whether E will occur; that is,
    - $V \in E$ and $V \notin E$ are considered *equally likely*.
  - How do we interpret other values of p?

Note: We could also define probabilities for more general propositions, as well as events.

Four Definitions of Probability

- Several alternative definitions of probability are commonly encountered:
  - Frequentist, Bayesian, Laplacian, Axiomatic
- They have different strengths & weaknesses, philosophically speaking.
  - But fortunately, they coincide with each other and work well together, in the majority of cases that are typically encountered.

Probability: Frequentist Definition

- The probability of an event E is the limit, as $n \to \infty$, of the fraction of times that we find $V \in E$ over the course of n independent repetitions of (different instances of) the same experiment.

$$\Pr[E] :\equiv \lim_{n\to\infty} \frac{n_{V\in E}}{n}$$

- Some problems with this definition:
  - It is only well-defined for experiments that can be independently repeated, infinitely many times!
    - or at least, if the experiment can be repeated in principle, *e.g.*, over some hypothetical ensemble of (say) alternate universes.
  - It can never be measured exactly in finite time!
- **Advantage:** It's an objective, mathematical definition.

Probability: Bayesian Definition

- Suppose a rational, profit-maximizing entity R is offered a choice between two rewards:
  - Winning \$1 if and only if the event E actually occurs.
  - Receiving p dollars (where $p \in [0,1]$) unconditionally.
- If R can honestly state that he is completely indifferent between these two rewards, then we say that R's probability for E is p, that is, $\Pr_R[E] :\equiv p$.
- **Problem:** It's a subjective definition; depends on the reasoner R, and his knowledge, beliefs, & rationality.
  - The version above additionally assumes that the utility of money is linear.
    - This assumption can be avoided by using "utils" (utility units) instead of dollars.

Probability: Laplacian Definition

- First, assume that all individual outcomes in the sample space are *equally likely* to each other…
- Then, the probability of any event E is given by,

  $$\Pr[E] = |E|/|S|.$$

- **Problems:** Still needs a definition for *equally likely*, and depends on the existence of *some* finite sample space S in which all outcomes in S are, in fact, equally likely.

Probability: Axiomatic Definition

- Let p be any total function $p{:}S\to[0,1]$ such that
$$\sum_s p(s) = 1.$$
- Such a p is called a *probability distribution*.
- Then, the *probability under p* of any event $E \subseteq S$ is just:
$$p[E] :\equiv \sum_{s\in E} p(s)$$
- **Advantage:** Totally mathematically well-defined!
  - This definition can even be extended to apply to infinite sample spaces, by changing $\sum\to\int$, and calling p a *probability density function* or a probability *measure*.
- **Problem:** Leaves operational meaning unspecified.

The Axioms of Probability

- $0 <= P(A) <= 1$
- $P(True) = 1$
- $P(False) = 0$
- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$

These Axioms are Not to be Trifled With

There have been attempts to do different methodologies for uncertainty

- Fuzzy Logic
- Three-valued logic
- Dempster-Shafer
- Non-monotonic reasoning

Probability of an event E

The probability of an event E is the sum of the probabilities of the outcomes in E. That is

$$p(E) = \sum_{s \in E} p(s)$$

Note that, if there are n outcomes in the event E, that is, if E = $\{a_1, a_2, \ldots, a_n\}$ then

$$p(E) = \sum_{i=1}^{n} p(a_i)$$

Example

What is the probability that, if we flip a coin three times, that we get an odd number of tails?

(TTT), (TTH), (THH), (HTT), (HHT), (HHH), (THT), (HTH)

Each outcome has probability 1/8,

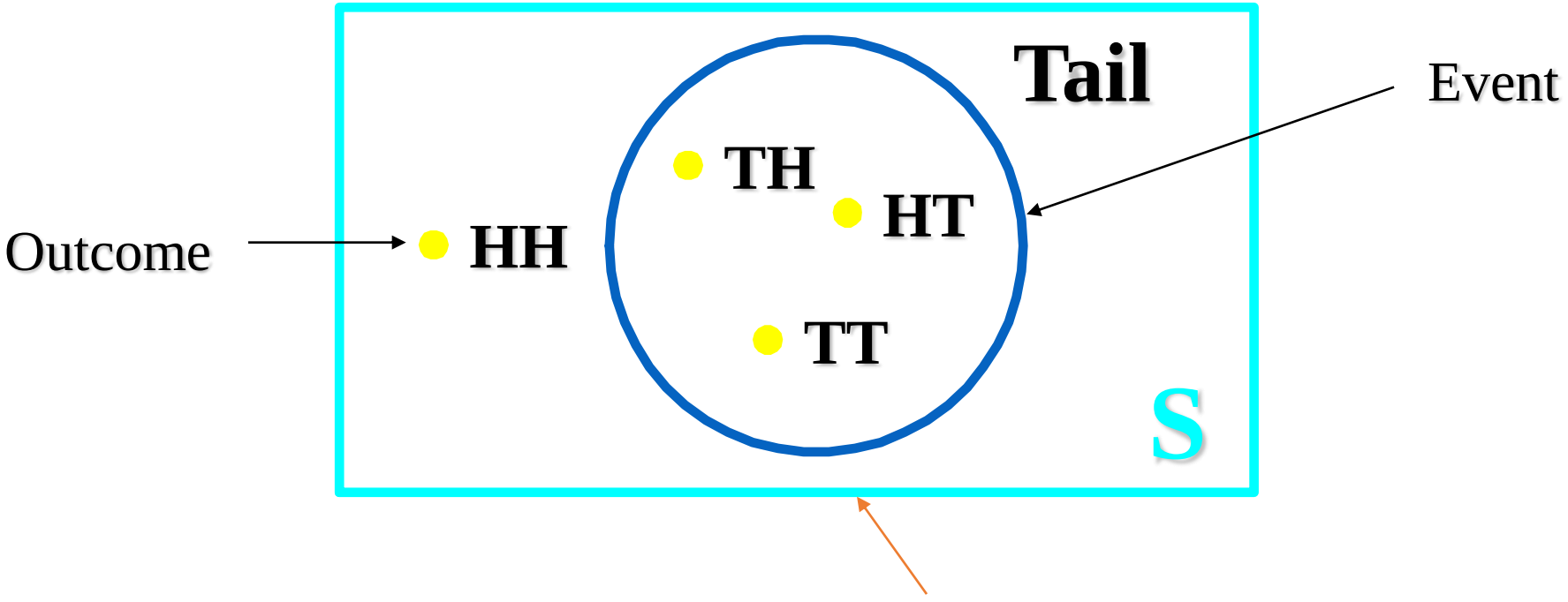
p(odd number of tails) = 1/8+1/8+1/8+1/8 = ½

Visualizing Sample Space1

- Listing : S = {Head, Tail}
- Venn Diagram
- Contingency Table
- Decision Tree Diagram

Venn Diagram

Experiment: Toss 2 Coins. Note Faces.



S = {HH, HT, TH, TT}

Contingency Table

Experiment: Toss 2 Coins. Note Faces.

1st Coin	2nd Coin: Head	2nd Coin: Tail	Total
Head	HH	HT	HH, HT
Tail	TH	TT	TH, TT
Total	HH, TH	HT, TT	S

Simple Event (Head on 1st Coin)

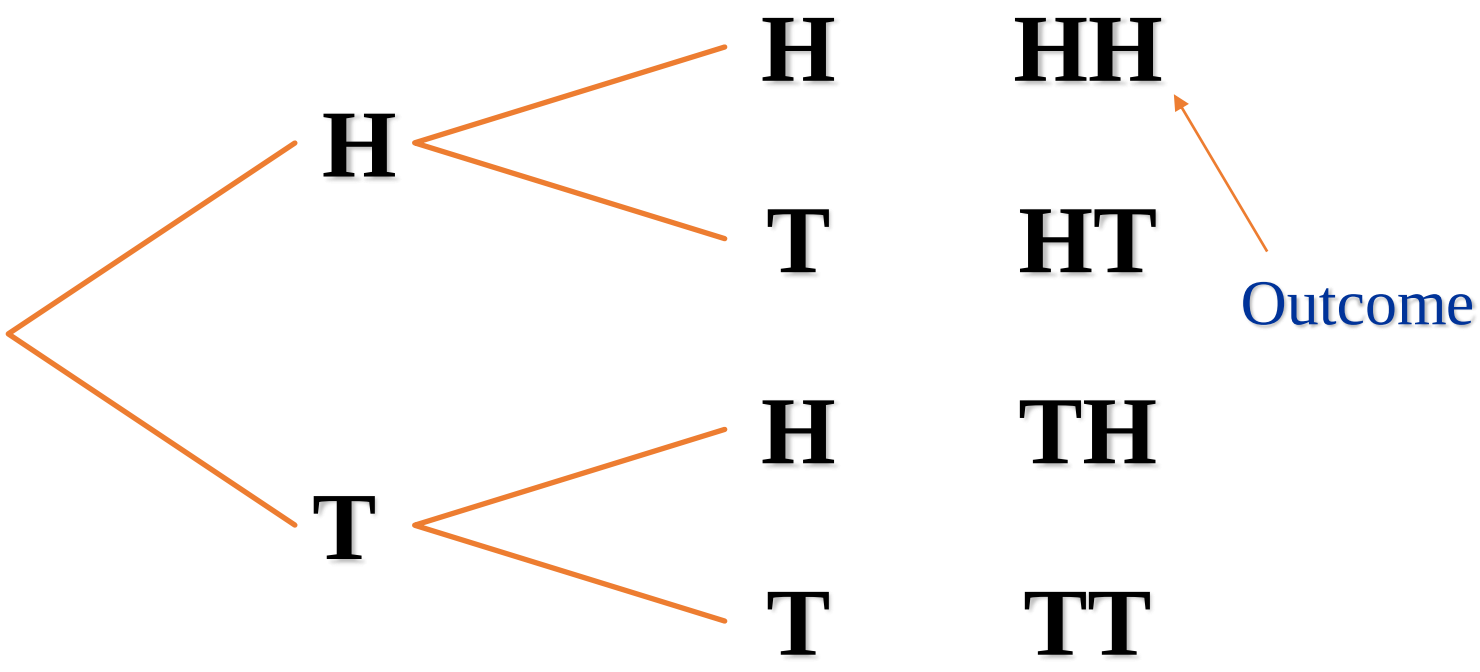
Outcome

Sample Space

S = {HH, HT, TH, TT}

Tree Diagram

Experiment: Toss 2 Coins. Note Faces.



S = {HH, HT, TH, TT}

Sample Space

Discrete Random Variable

- Possible values (outcomes) are discrete
  - E.g., natural number (0, 1, 2, 3 etc.)
- Obtained by Counting
- Usually Finite Number of Values
  - But could be infinite (must be “countable”)

Discrete Probability Distribution
(also called probability mass function (pmf))

1. List of All possible [x, $p(x)$] pairs
   - x = Value of Random Variable (Outcome)
   - $p(x)$ = Probability Associated with Value
2. Mutually Exclusive (No Overlap)
3. Collectively Exhaustive (Nothing Left Out)
4. $0 \leq p(x) \leq 1$
5. $\sum p(x) = 1$

Visualizing Discrete Probability Distributions

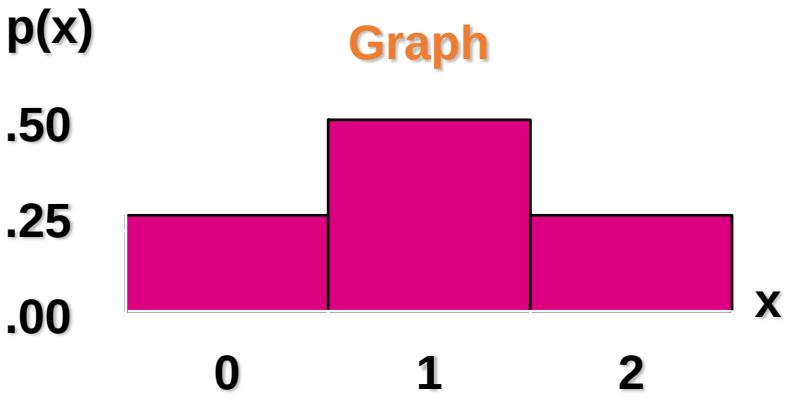
Listing

{ (0, .25), (1, .50), (2, .25) }

Table

# Tails	f(x) Count	p(x)
0	1	.25
1	2	.50
2	1	.25

Graph



Equation

$$p(x) = \frac{n!}{x!(n-x)!} p^x (1-p)^{n-x}$$

Arity of Random Variables

- Suppose A can take on more than 2 values
- A is a *random variable with arity k* if it can take on exactly one value out of $\{v_1, v_2, .. v_k\}$
- Thus…

$$P(A = v_i \wedge A = v_j) = 0 \text{ if } i \neq j$$
$$P(A = v_1 \vee A = v_2 \vee A = v_k) = 1$$

Mutually Exclusive Events

- Two events E_1, E_2 are called *mutually exclusive* if they are disjoint:

$$E_1 \cap E_2 = \varnothing$$

Note that two mutually exclusive events *cannot both occur* in the same instance of a given experiment.

- For mutually exclusive events,

$$\Pr[E_1 \cup E_2] = \Pr[E_1] + \Pr[E_2].$$

Exhaustive Sets of Events

- A set $\boldsymbol{E} = \{E_1, E_2, \ldots\}$ of events in the sample space S is called *exhaustive* iff

$$\bigcup E_i = S$$

- An exhaustive set $\boldsymbol{E}$ of events that are all mutually exclusive with each other has the property that

$$\sum \Pr[E_i] = 1.$$

Bernoulli Trials

- Each performance of an experiment with only two possible outcomes is called a **Bernoulli trial**.
- In general, a possible outcome of a Bernoulli trial is called a **success** or a **failure**.
- If p is the probability of a success and q is the probability of a failure, then $p+q=1$.

Example

A coin is biased so that the probability of heads is 2/3. What is the probability that exactly four heads come up when the coin is flipped seven times, assuming that the flips are independent?

The number of ways that we can get four heads is:

$C(7,4) = 7!/4!3! = 7*5 = 35$

The probability of getting four heads and three tails is

$(2/3)^4(1/3)^{3=} 16/3^7$

p(4 heads and 3 tails) is

$C(7,4)\ (2/3)^4(1/3)^3 = 35*16/3^7 = 560/2187$

Probability of k successes in n independent Bernoulli trials.

The probability of k successes in n independent Bernoulli trials, with probability of success p and probability of failure q = 1-p is $C(n,k)p^k q^{n-k}$

Find each of the following probabilities when n independent Bernoulli trials are carried out with probability of success, p.

- Probability of no successes.

$$C(n,0)p^0q^{n-k} = 1(p^0)(1-p)^n = (1-p)^n$$

- Probability of at least one success.

$$1 - (1-p)^n$$

Find each of the following probabilities when n independent Bernoulli trials are carried out with probability of success, p.

- Probability of at most one success.

  Means there can be no successes or one success

  $C(n,0)p^0q^{n-0} + C(n,1)p^1q^{n-1}$

  $(1-p)^n + np(1-p)^{n-1}$

- Probability of at least two successes.

  $1 - (1-p)^n - np(1-p)^{n-1}$

A coin is flipped until it comes ups tails. The probability the coin comes up tails is p.

- What is the probability that the experiment ends after n flips, that is, the outcome consists of n-1 heads and a tail?

$(1-p)^{n-1}p$

Probability vs. Odds

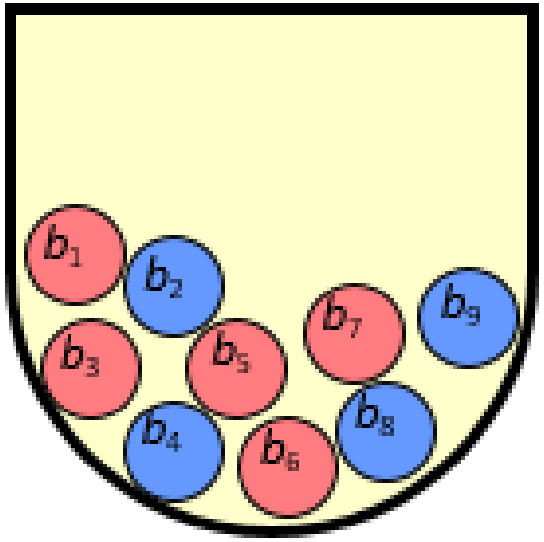
- You may have heard the term "odds."
  - It is widely used in the gambling community.
- This is not the same thing as probability!
  - But, it is very closely related.
- The *odds in favor* of an event E means the *relative* probability of $\overline{E}$ compared with its complement E.

$$O(E) :\equiv \Pr(E)/\Pr(\overline{E}).$$

  - *E.g.*, if $p(E) = 0.6$ then $p(\overline{E}) = 0.4$ and $O(E) = 0.6/0.4 = 1.5$.
- Odds are conventionally written as a ratio of integers.
  - *E.g.*, 3/2 or 3:2 in above example. "Three to two in favor."
- The *odds against E* just means $1/O(E)$. "2 to 3 against"

Example 1: Balls-and-Urn

- Suppose an urn contains 4 blue balls and 5 red balls.
- An example **experiment:** Shake up the urn, reach in (without looking) and pull out a ball.
- A **random variable** V: Identity of the chosen ball.
- The **sample space** S: The set of all possible values of V:
  - In this case, $S = \{b_1,\ldots,b_9\}$
- An **event** E: "The ball chosen is blue": $E = \{$ _____ $\}$
- What are the odds in favor of E?
- What is the probability of E?

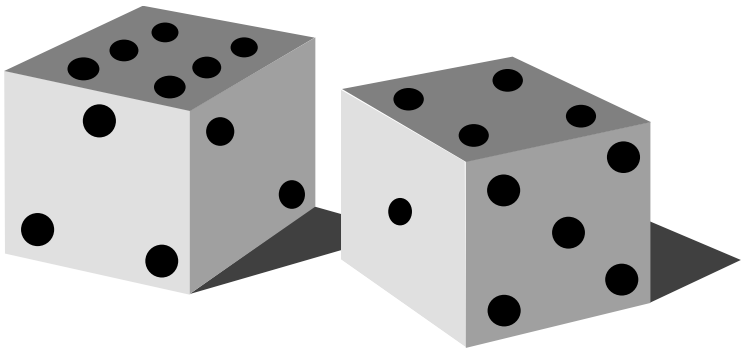


Independent Events

- Two events E,F are called *independent* if
  $$\Pr[E \cap F] = \Pr[E] \cdot \Pr[F].$$
- Relates to the product rule for the number of ways of doing two independent tasks.
- **Example:** Flip a coin, and roll a die.
  $$\Pr[(\text{coin shows heads}) \cap (\text{die shows } 1)] =$$
  $$\Pr[\text{coin is heads}] \times \Pr[\text{die is } 1] = \tfrac{1}{2} \times 1/6 = 1/12.$$

Example

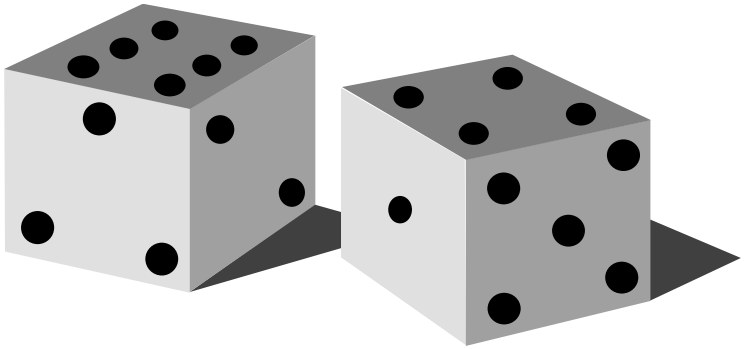
Suppose a red die and a blue die are rolled. The sample space:



	1	2	3	4	5	6
1	x	x	x	x	x	x
2	x	x	x	x	x	x
3	x	x	x	x	x	x
4	x	x	x	x	x	x
5	x	x	x	x	x	x
6	x	x	x	x	x	x

Are the events sum is 7 and the blue die is 3 independent?

The events sum is 7 and
the blue die is 3 are independent:

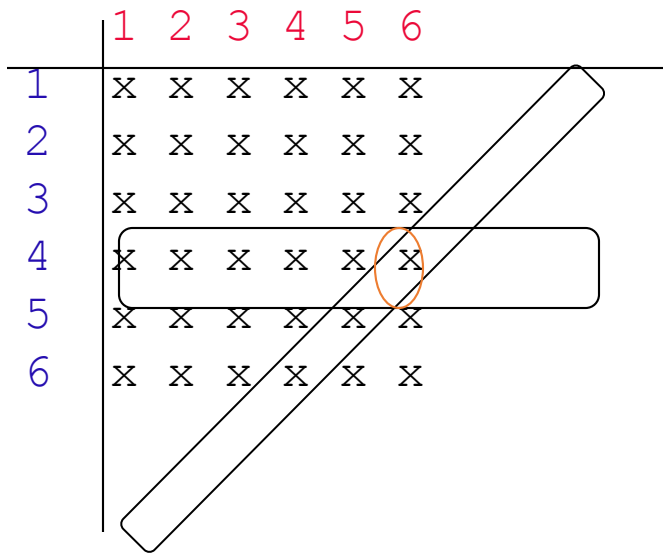


$|S| = 36$

$|\text{sum is } 7| = 6$

$|\text{blue die is } 3| = 6$

$|\text{ in intersection }| = 1$



p(sum is 7 and blue die is 3) =1/36

p(sum is 7) p(blue die is 3) =6/36*6/36=1/36

Thus, p((sum is 7) and (blue die is 3)) = p(sum is 7) p(blue die is 3)

Conditional Probability

- Let E, F be any events such that $\Pr[F] > 0$.
- Then, the *conditional probability of* E *given* F, written $\Pr[E|F]$, is defined as

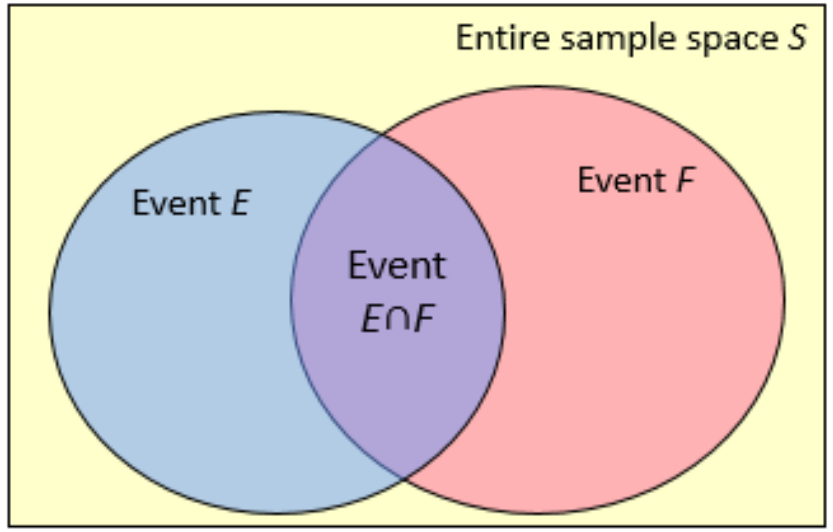
$$\Pr[E|F] :\equiv \Pr[E \cap F]/\Pr[F].$$

- This is what our probability that E would turn out to occur should be, if we are given *only* the information that F occurs.
- If E and F are independent then $\Pr[E|F] = \Pr[E]$.

$$\because \Pr[E|F] = \Pr[E \cap F]/\Pr[F] = \Pr[E] \times \Pr[F]/\Pr[F] = \Pr[E]$$

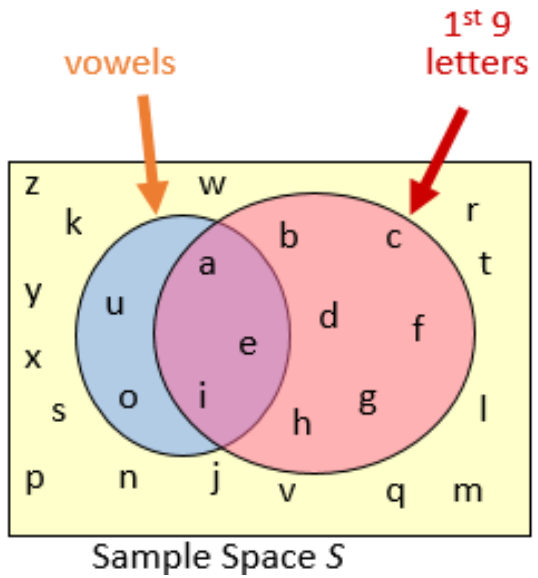
Visualizing Conditional Probability

- If we are given that event F occurs, then
  - Our attention gets restricted to the subspace F.
- Our *posterior* probability for E (after seeing F) corresponds to the *fraction* of F where E occurs also.
- Thus, $p'(E) = p(E \cap F)/p(F)$



Conditional Probability Example

- Suppose I choose a single letter out of the 26-letter English alphabet, totally at random.
  - Use the Laplacian assumption on the sample space {a,b,..,z}.
  - What is the (prior) probability that the letter is a vowel?
    - Pr[Vowel] = __ / __ .
- Now, suppose I tell you that the letter chosen happened to be in the first 9 letters of the alphabet.
  - Now, what is the conditional (or posterior) probability that the letter is a vowel, given this information?
    - Pr[Vowel | First9] = ___ / ___ .



Example

- What is the probability that, if we flip a coin three times, that we get an odd number of tails (=event E), if we know that the event F, the first flip comes up tails occurs?

(TTT), (TTH), (THH), (HTT),
(HHT), (HHH), (THT), (HTH)

Each outcome has probability 1/4,

p(E |F) = 1/4+1/4 = ½, where E=odd number of tails
or p(E|F) = p($E \cap F$)/p(F) = 2/4 = ½
For comparison p(E) = 4/8 = ½
E and F are independent, since p(E |F) = Pr(E).

Prior and Posterior Probability

- Suppose that, before you are given any information about the outcome of an experiment, your personal probability for an event E to occur is $p(E) = \Pr[E]$.
  - The probability of E in your original probability distribution p is called the *prior* probability of E.
    - This is its probability *prior* to obtaining any information about the outcome.
- Now, suppose someone tells you that some event F (which may overlap with E) actually occurred in the experiment.
  - Then, you should *update* your personal probability for event E to occur, to become $p'(E) = \Pr[E|F] = p(E \cap F)/p(F)$.
    - The conditional probability of E, *given* F.
  - The probability of E in your *new* probability distribution p' is called the *posterior* probability of E.
    - This is its probability *after* learning that event F occurred.
- After seeing F, the posterior distribution p' is defined by letting $p'(v) = p(\{v\} \cap F)/p(F)$ for each individual outcome $v \in S$.